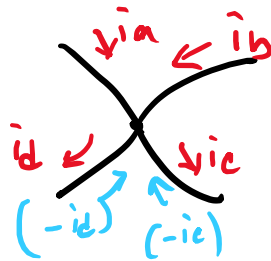


Kirchoff's laws

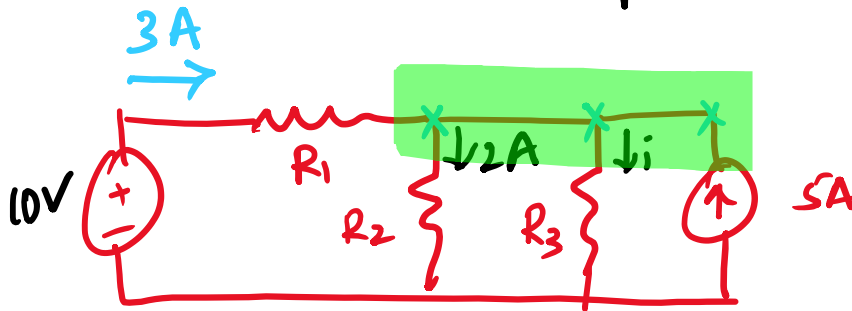
- Current law
- voltage law.

→ KCL : The algebraic sum of currents entering any node is zero.

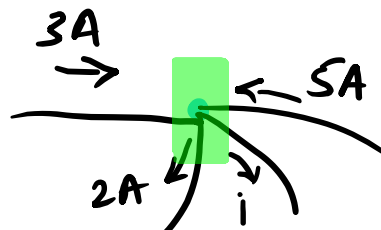
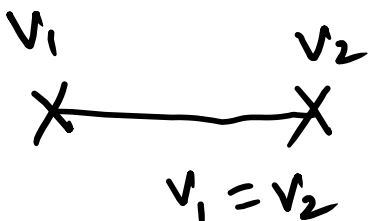


$$i_a + i_b + (-i_c) + (-i_d) = 0$$

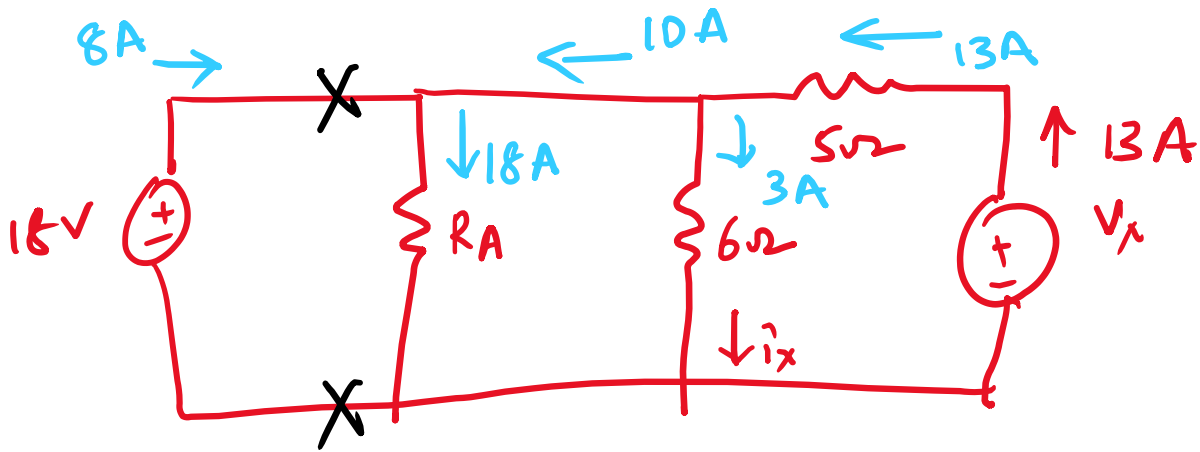
Total incoming = total outgoing current at any node.



If the current from the 10V source is 3A, find the current through R_3

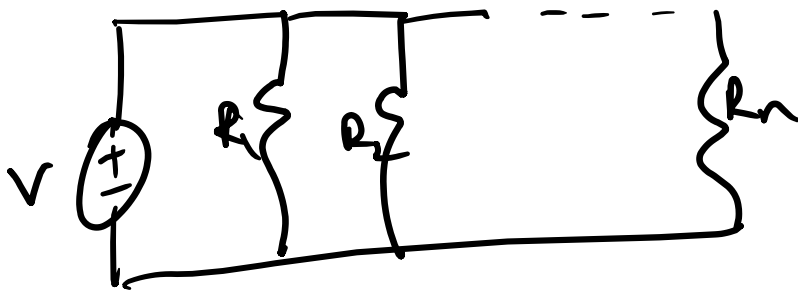


$$5 + 3 = 2 + i$$
$$i = \underline{6A}$$



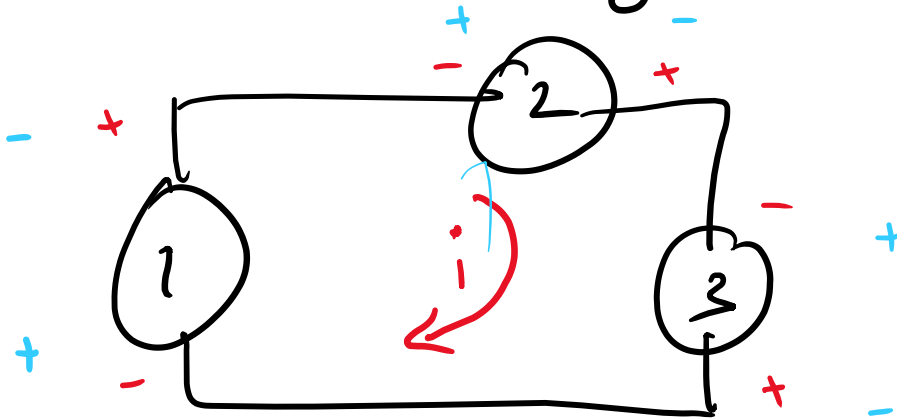
If $i_x = 3A$, 18V source delivers a current of 8A, what is the value of R_A

$$V_{RA} = 18V \quad \hat{i}_{RA} = 18A \quad R_A = \underline{1\Omega}$$



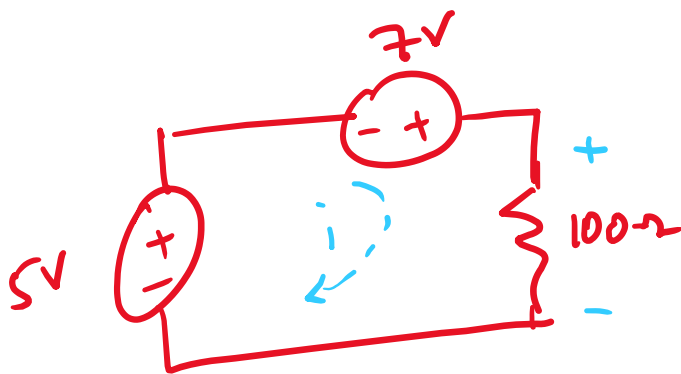
$$\begin{aligned} V_{R_1} &= V_{R_2} \\ &= \dots = V_{R_n} \\ &= V \end{aligned}$$

KVL: Algebraic sum of voltages in a closed loop is zero.



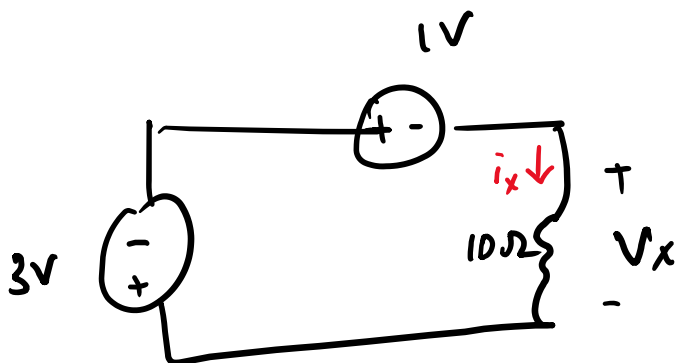
$$V_1 + V_2 + V_3 = 0$$

$$-V_1 - V_2 - V_3 = 0$$



Find V_x using KVL

$$+5 + 7 - V_x = 0 \Rightarrow V_x = 12V$$

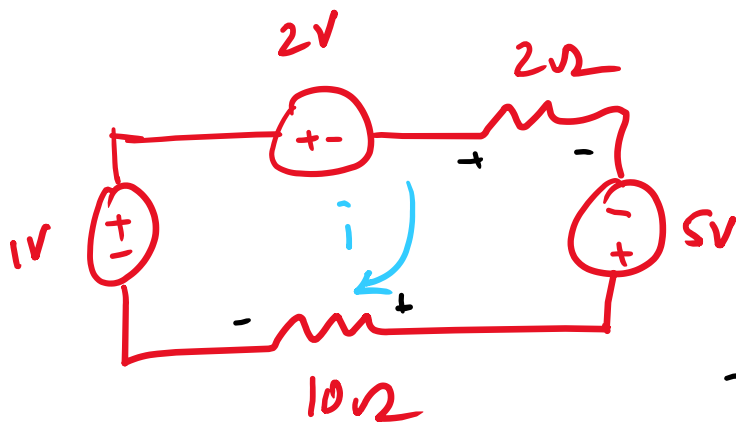


Find V_x, i_x

$$-3V - 1V - V_x = 0$$

$$V_x = -4V$$

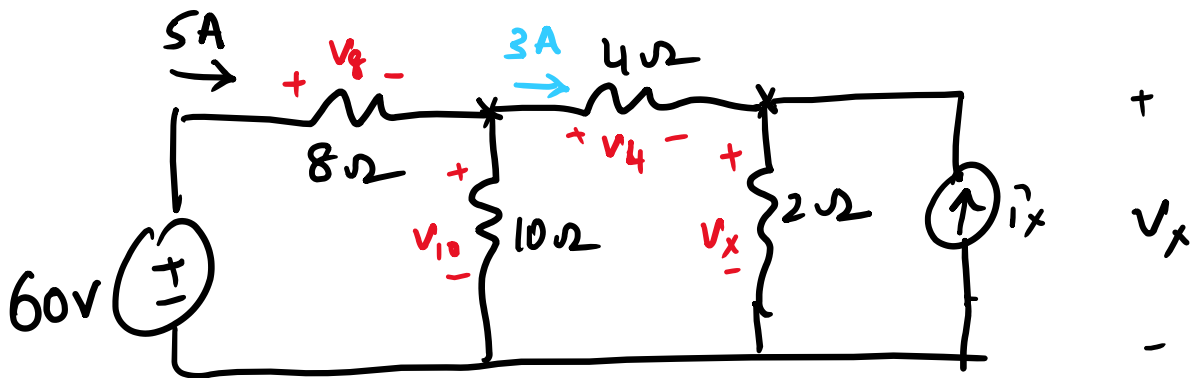
$$i_x = \frac{-4V}{10} = -0.4A$$



Find i

$$+1 - 2 - 2i + 5 - 10i = 0$$

$$i = \underline{0.33A}$$



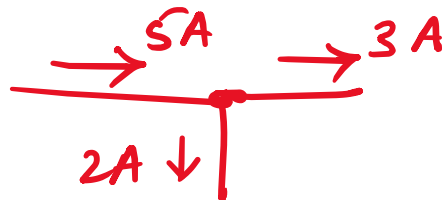
Find i_x, v_x

$$v_8 = 5 \times 8 = 40V$$

KVL

$$+60 - 40 - v_{10} = 0 \Rightarrow v_{10} = 20V$$

$$i_{10} = \frac{v_{10}}{10} = 2A$$



$$v_4 = 3 \times 4 = 12V$$

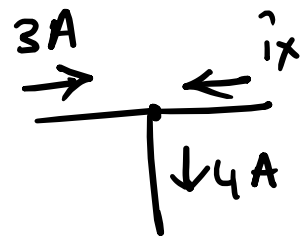


$$v_{10} - v_4 - v_x = 0$$

$$20 - 12 - v_x = 0$$

$$v_x = 8V$$

$$\hat{i}_2 = \frac{V_x}{2\Omega} = 4A$$



$$\underline{\hat{i}_x = 1A}$$